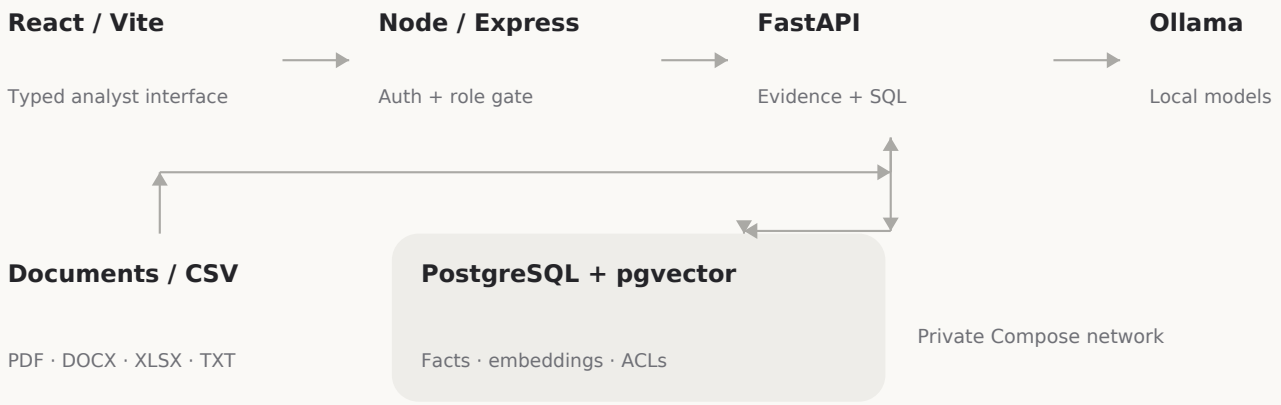
# Ollama Private Business Intelligence

Verified local stack · Self-hosted inference · Financial SQL + permission-filtered document RAG

## 1. Execution topology




## Authority and model boundaries

| Responsibility | Implementation / enforced contract |
|---|---|
| Financial metrics | Fixed, parameterized PostgreSQL queries and Decimal arithmetic. No model-generated SQL or model-authored financial figures. |
| Document evidence | PDF, DOCX, XLSX, CSV and TXT extraction; local nomic-embed-text vectors (768 dimensions), stored and searched with pgvector. |
| Local inference | Ollama qwen2.5:1.5b selects validated evidence IDs; rendered answers use authoritative calculations and exact stored excerpts. |
| Access scope | Express authenticates, authorizes and passes scoped identity to an internal-token-protected FastAPI runtime; document ACL checks occur at retrieval/read time. |

## 2. Selected API contracts

| Route | Purpose |
|---|---|
| POST /api/analysis | Route a question to metrics, retrieval or hybrid analysis; return findings, metrics, source excerpts and warnings. |
| POST /api/documents | Upload and index supported local files; retain durable processing state and error/retry information. |
| POST /api/sales/import | Validated, idempotent customer/sales CSV import; no partial writes on rejected rows. |
| POST /api/expenses/import | Validate expense CSV including optional existing-customer attribution. |
| GET /api/analyses | Restore private saved analyses, rechecking access to every supporting document. |

# What the local audit establishes

Evidence from the repository completion audit dated 2 October 2026. These are demonstration results, not production performance benchmarks.

## 3. Recorded verification

| Verification layer | Observed result |
|---|---|
| Real local inference | 11 business questions with local qwen2.5:1.5b; successful real nomic-embed-text embeddings and 768-dimensional vectors. |
| Business + isolation audit | 82 assertions against a synthetic organisation: independent SQL checks, ACLs, five extraction formats, retries, imports and history. |
| Unit and browser | 13 Python tests; 3 Node tests plus frontend type check; 2 Chromium end-to-end journeys; backend/frontend builds passed. |
| Runtime and recovery | Five Compose services running, readiness HTTP 200, all eight SQL migrations, model-missing/recovery test (6 assertions). |

## 4. Synthetic business fixture (2027)

| Question | Verified recorded result |
|---|---|
| Q1 to Q2 revenue | ZAR 200,000 → ZAR 120,000; change −ZAR 80,000 (−40%). |
| Q2 recorded company profit | ZAR 120,000 revenue minus ZAR 137,000 recorded expenses = −ZAR 17,000. |
| Customer gross profit | Synthetic Beta ZAR 45,000; Synthetic Alpha ZAR 110,000. Unattributed overhead excluded. |
| Reported causes | Management excerpts describe supplier delays and order-processing disruption; not verified causation or loss attribution. |

## 5. Security and failure boundaries

| Control | Scope and limitation |
|---|---|
| Session lifecycle | 30-minute JWTs in browser memory; rotating, hashed refresh sessions in HttpOnly cookies. Logout, replay and disabling an account revoke access. |
| Source visibility | Document access restrictions for uploader, administrators, users or roles; scope enforced in search, metadata and saved-analysis reads. |
| Private deployment | Compose services bind localhost; AI runtime is not host-published. Model/image downloads use network access during setup; not certified air-gapped. |
| Production limits | No verified public deployment. Needs TLS, secrets management, dedicated read-only database privileges, encrypted backups, durable job recovery and security assessment. |

SOURCE / github.com/keenosmith-del/ollama-private-business-intelligence · docs/COMPLETION_AUDIT.md